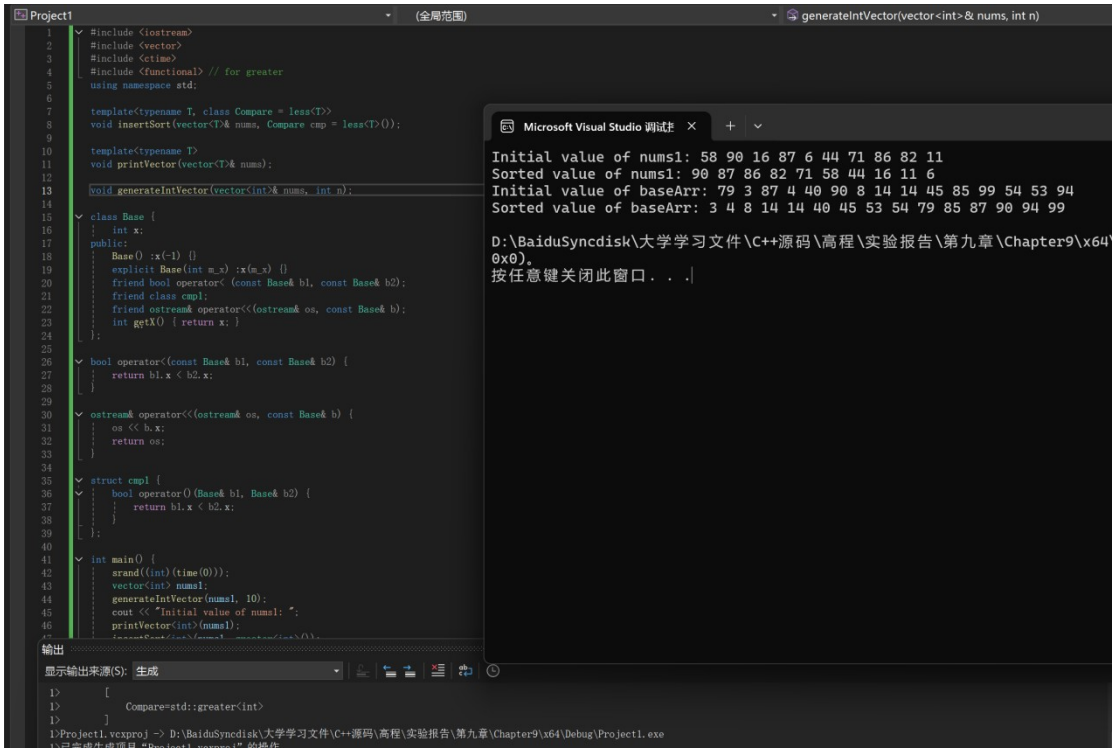
第九章---模板和群体数据

例题



练习 1

Node.h | Link.h | Node.h | lab9-4.cpp | array1.h | lab9-

Project2 (全局范围)

```cpp
    void clear();
};

template<typename T>
LinkedList<T>::~LinkedList() {
    clear();
}

template<typename T>
void LinkedList<T>::insertAtEnd(const T& item) {
    Node<T>* newNode = new Node<T>(item);
    if (!head) {
        head = newNode;
    }
    else {
        Node<T>* temp = head;
        while (temp->getNext()) {
            temp = temp->getNext();
        }
        temp->setNext(newNode);
    }
}

template<typename T>
void LinkedList<T>::display() const {
    Node<T>* temp = head;
    while (temp) {
        cout << temp->getData() << " ";
        temp = temp->getNext();
    }
    cout << endl;
}

template<typename T>
bool LinkedList<T>::search(const T& key) const {
    Node<T>* temp = head;
    while (temp) {
        if (temp->getData() == key) {
            return true;
        }
        temp = temp->getNext();
    }
    return false;
}

template<typename T>
void LinkedList<T>::remove(const T& key) {
    while (head && head->getData() == key) {
        Node<T>* toDelete = head;
        head = head->getNext();
        delete toDelete;
    }
    if (!head) return;

    Node<T>* current = head;
    while (current->getNext()) {
        if (current->getNext()->getData() == key) {
```

```
5 6 3 4 7 8 9 1 2 5
Linked list contents: 5 6 3 4 7 8 9 1 2 5
Enter a value to search and delete: 3
After deletion, list contents: 5 6 4 7 8 9 1 2 5

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Project2.exe (进程 7532)已退出，代码为 0 (0x0)。
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```

练习 2

Project3 (全局范围)

```cpp
public:
    LinkedList() : head(nullptr) {}
    ~LinkedList();
    void insertAtEnd(const T& item);
    void display() const;
    void concatenate(LinkedList<T>& other);
};

template<typename T>
LinkedList<T>::~LinkedList() {
    Node<T>* current = head;
    while (current) {
        Node<T>* next = current->getNext();
        delete current;
        current = next;
    }
}

template<typename T>
void LinkedList<T>::insertAtEnd(const T& item) {
    Node<T>* newNode = new Node<T>(item);
    if (!head) {
        head = newNode;
    }
    else {
        Node<T>* temp = head;
        while (temp->getNext()) {
            temp = temp->getNext();
        }
        temp->setNext(newNode);
    }
}

template<typename T>
void LinkedList<T>::display() const {
    Node<T>* temp = head;
    while (temp) {
        cout << temp->getData() << " ";
        temp = temp->getNext();
    }
    cout << endl;
}

template<typename T>
void LinkedList<T>::concatenate(LinkedList<T>& other) {
    if (!head) {
        head = other.head;
    }
    else {
        Node<T>* temp = head;
        while (temp->getNext()) {
            temp = temp->getNext();
        }
        temp->setNext(other.head);
    }
    other.head = nullptr; // 防止其他链表被析构时释放节点
}
```

Microsoft Visual Studio 调试控

```
Enter 5 integers for list A:
1 5 7 3 4
Enter 5 integers for list B:
2 5 9 8 7
List A: 1 5 7 3 4
List B: 2 5 9 8 7
After concatenation, List A: 1 5 7 3 4 2 5 9 8 7
List B (should be empty):

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0x0)。
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```
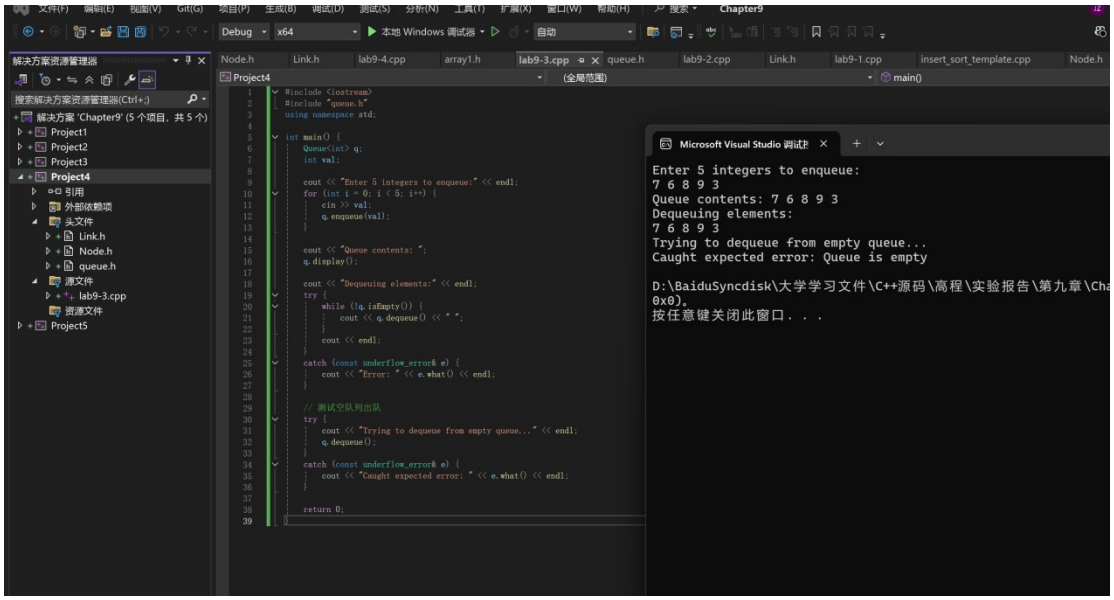

练习 3


Node.h
Link.h
lab9-4.cpp
array1.h
lab9-3.cpp
queue.h
lab9-2.cpp
Link.h
lab9-1.cpp
insert_sort_template.cpp
Node.h
Project4
(全局范围)
main()
#include <iostream>
#include "queue.h"
using namespace std;
int main() {
Queue<int> q;
int val;
cout << "Enter 5 integers to enqueue:" << endl;
for (int i = 0; i < 5; i++) {
cin >> val;
q.enqueue(val);
}
cout << "Queue contents: ";
q.display();
cout << "Dequeuing elements:" << endl;
try {
while (!q.isEmpty()) {
cout << q.dequeue() << " ";
}
cout << endl;
}
catch (const underflow_error& e) {
cout << "Error: " << e.what() << endl;
}
// 测试空队列出队
try {
cout << "Trying to dequeue from empty queue..." << endl;
q.dequeue();
}
catch (const underflow_error& e) {
cout << "Caught expected error: " << e.what() << endl;
}
return 0;
}
Microsoft Visual Studio 调试
Enter 5 integers to enqueue:
7 6 8 9 3
Queue contents: 7 6 8 9 3
Dequeuing elements:
7 6 8 9 3
Trying to dequeue from empty queue...
Caught expected error: Queue is empty
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0x0)。
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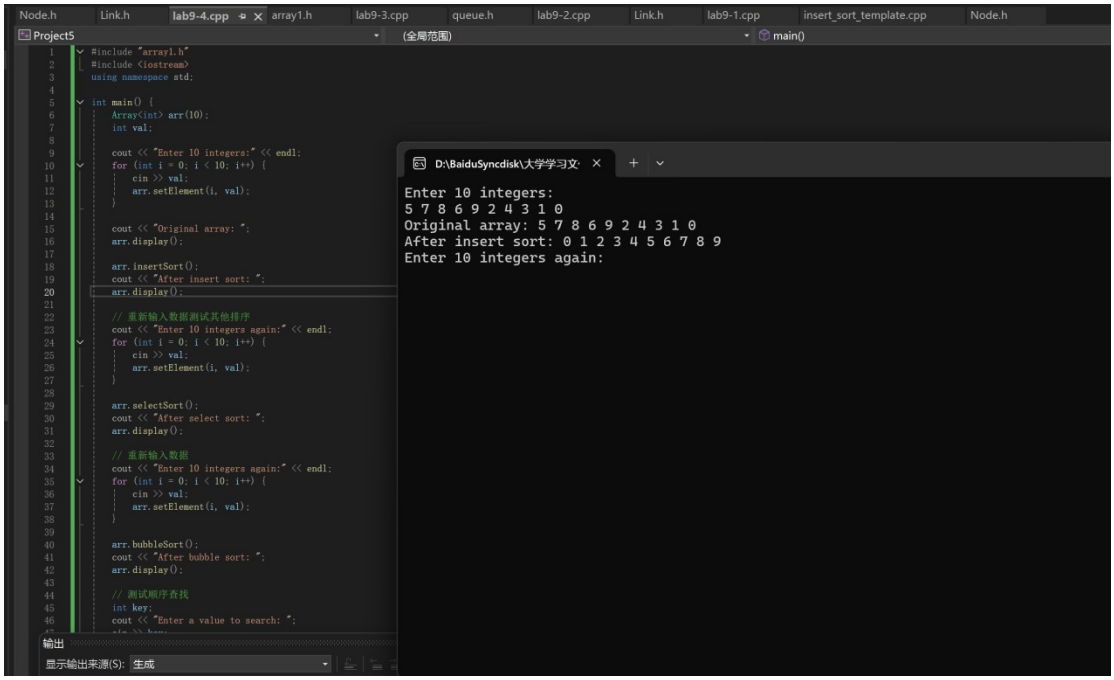
练习 4


Node.h
Link.h
lab9-4.cpp
array1.h
lab9-3.cpp
queue.h
lab9-2.cpp
Link.h
lab9-1.cpp
insert_sort_template.cpp
Node.h
Project5
(全局范围)
main()
#include "array1.h"
#include <iostream>
using namespace std;
int main() {
Array<int> arr(10);
int val;
cout << "Enter 10 integers:" << endl;
for (int i = 0; i < 10; i++) {
cin >> val;
arr.setElement(i, val);
}
cout << "Original array: ";
arr.display();
arr.insertSort();
cout << "After insert sort: ";
arr.display();
// 重新输入数据测试其他排序
cout << "Enter 10 integers again:" << endl;
for (int i = 0; i < 10; i++) {
cin >> val;
arr.setElement(i, val);
}
arr.selectSort();
cout << "After select sort: ";
arr.display();
// 重新输入数据
cout << "Enter 10 integers again:" << endl;
for (int i = 0; i < 10; i++) {
cin >> val;
arr.setElement(i, val);
}
arr.bubbleSort();
cout << "After bubble sort: ";
arr.display();
// 测试顺序查找
int key;
cout << "Enter a value to search: ";
输出
显示输出来源(S): 生成
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Enter 10 integers:
5 7 8 6 9 2 4 3 1 0
Original array: 5 7 8 6 9 2 4 3 1 0
After insert sort: 0 1 2 3 4 5 6 7 8 9
Enter 10 integers again: